KOÇ
UNIVERSITY
COLLEGE OF ENGINEERING

MECH 303

Machine Design

Project 2:

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10.12.2025

1. Introduction

Deep-sea exploration requires robust structural designs capable of withstanding extreme hydrostatic pressures, a critical engineering challenge recently highlighted by the OceanGate Titan incident, which serves as the reference case for this project. The primary objective of this study is to evaluate the structural integrity of a pressure hull composed of a Carbon Fiber Composite cylinder and Titanium hemispherical caps by utilizing Finite Element Analysis in ANSYS Workbench. The scope of the project includes performing a static structural analysis at depths ranging from 2000 m to 4500 m to determine stress distributions and safety factors, conducting a parametric study to quantify the effect of wall thickness on safety margins, and executing an elasto-plastic analysis to identify the yield pressure and observe residual plastic strains, thereby providing a comprehensive understanding of the design limits and failure mechanics of deep-sea submersibles.

2. Problem Description

The primary objective of this project is to analyze the structural integrity of a deep-sea submersible pressure hull under hydrostatic pressure conditions. The geometry of the pressure vessel is modeled based on the "Titan" submersible design, consisting of a cylindrical shell body and two hemispherical endcaps[1]. The study aims to compute membrane stresses, evaluate safety factors, and investigate the effects of wall thickness and material nonlinearity (plasticity) on the structural performance of the vessel.

2.1. Geometry and Model Details

The pressure hull geometry comprises a central cylinder made of Carbon Fiber Composite and two hemispherical caps made of Titanium Alloy (Ti-6Al-4V). The dimensions are defined with an inner cylinder radius (R_{ic}) of 355 mm and a wall thickness (T_c) of 63.5 mm. The hemispherical caps share the same inner radius and thickness to ensure geometric continuity [2]. Due to the symmetry of the geometry and loading, the model is simplified using symmetry planes to reduce computational cost while maintaining accuracy.

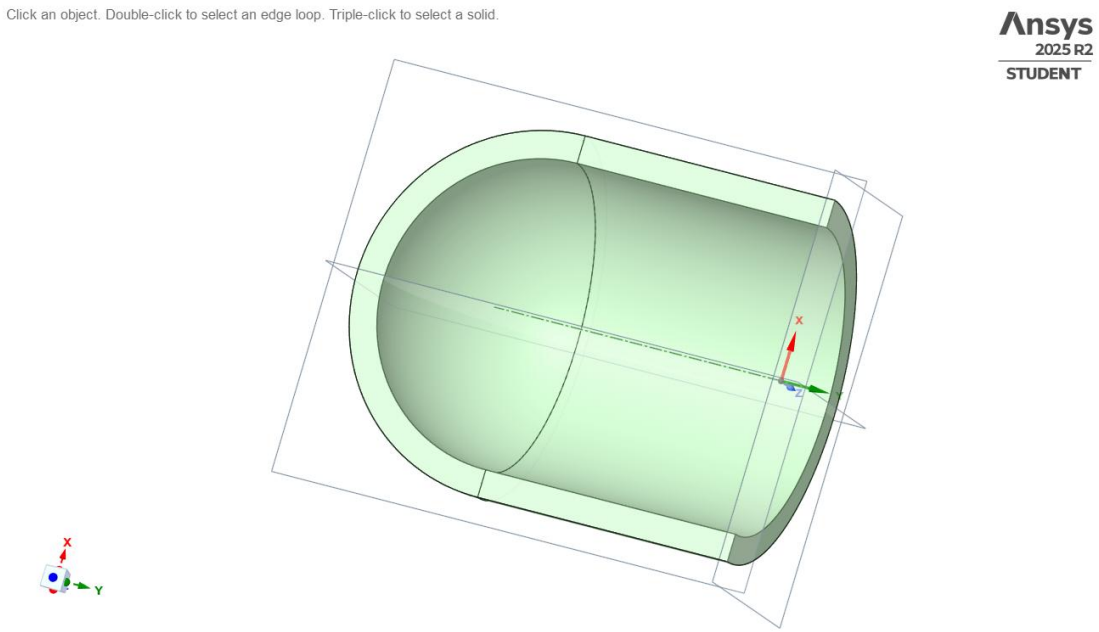


Figure 1: 3D Geometry model of the pressure vessel showing the Cylinder and Hemispherical Caps with applied symmetry cuts.

2.2. Material Properties

The structural components are assigned distinct material properties to simulate the multi-material design of the submersible. The cylindrical section is modeled as an equivalent isotropic Carbon Fiber Composite, while the endcaps are modeled using Ti-6Al-4V. For the initial static analysis, linear elastic properties are used. In the subsequent elasto-plastic analysis (Task 3), a Bilinear Isotropic Hardening model is applied to the Titanium caps to simulate post-yield behavior. The material properties used in this analysis are summarized in Table 1 below

Properties of Outline Row 5: Titanium			
	A	B	C
1	Property	Value	Unit
2	Material Field Variables	Table	
3	Density	4430	kg m^-3
4	Isotropic Elasticity		
5	Derive from	Young's Modulus an...	
6	Young's Modulus	110	GPa
7	Poisson's Ratio	0,34	
8	Bulk Modulus	1,1458E+11	Pa
9	Shear Modulus	4,1045E+10	Pa
10	Tensile Yield Strength	880	MPa

Properties of Outline Row 3: carbon fiber			
	A	B	C
1	Property	Value	Unit
2	Material Field Variables	Table	
3	Density	1600	kg m^-3
4	Isotropic Elasticity		
5	Derive from	Young's Modulus an...	
6	Young's Modulus	80	GPa
7	Poisson's Ratio	0,25	
8	Bulk Modulus	5,3333E+10	Pa
9	Shear Modulus	3,2E+10	Pa
10	Tensile Yield Strength	600	MPa

Table 1: Material Properties of Titanium and carbon fiber that is used in parts 1 and 2

2.3. Loading and Boundary Conditions

The submersible is subjected to external hydrostatic pressure, which varies as a function of depth. The applied pressure (P) is calculated using the formula

$$P = \rho g h$$

[1], where ρ is the seawater density ($1025 \sim kg/m^3$), g is the gravitational acceleration ($9.81 \sim m/s^2$), and h is the depth [4].

The Finite Element Model (FEM) is set up with the following boundary conditions: Displacement constraints are applied to the cut faces to represent the full 3D geometry. A "Bonded" contact type is defined between the cylinder and the caps to simulate a rigid connection. To prevent rigid body motion, displacement constraints are applied to the symmetry planes, and a single vertex is fixed.

The geometry is meshed using Tetrahedral elements with an element size 25 mm to ensure mesh quality and result convergence.

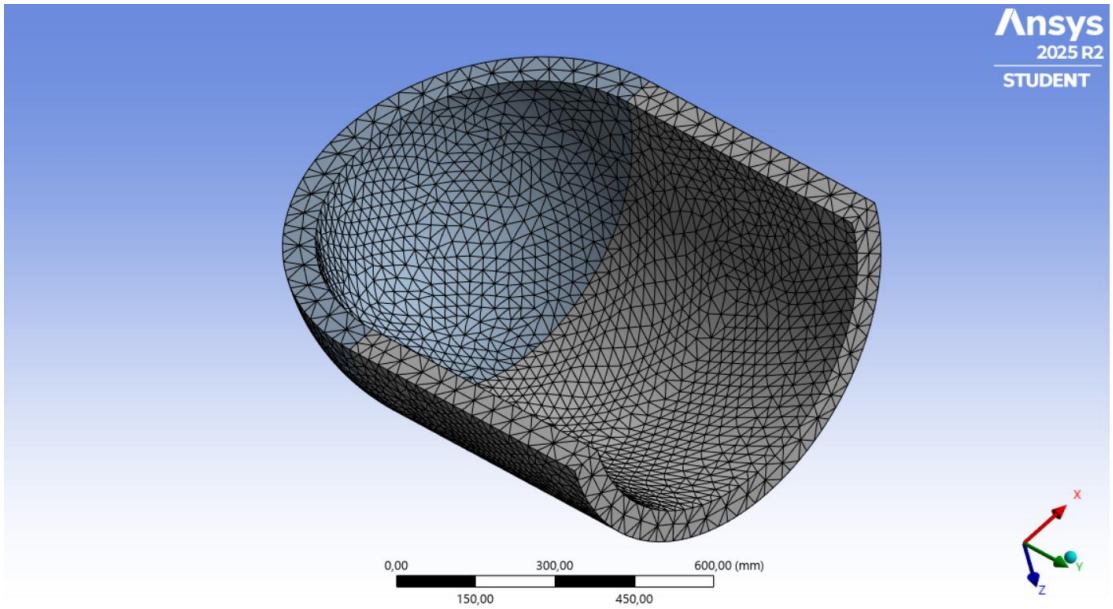


Figure 2: Finite Element Mesh of the assembly using Tetrahedral elements with a size of 25 mm.

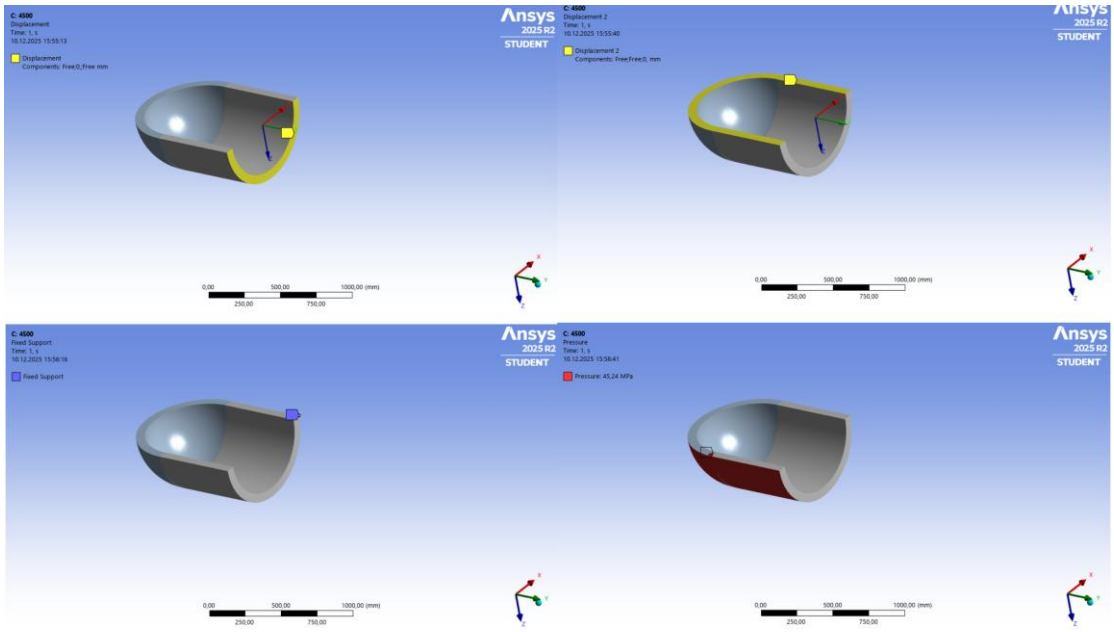


Figure 3: Boundary Conditions setup: showing Symmetry Regions, Fixed Support, and Applied Pressure Area.

2.4. Analysis Scope

The project is divided into three main tasks:

- **Task 1:** A static structural analysis is performed at three distinct depths (2000 m, 3800 m, and 4500 m) to determine the Von-Mises stress, directional stresses (axial and hoop), and safety factors.
- **Task 2:** A parametric study is conducted to evaluate the effect of wall thickness (ranging from 40 mm to 100 mm) on the minimum safety factors of the cylinder and the cap at a constant depth of 3800 m.
- **Task 3:** Elastic-plastic analysis is performed using a load-unload step definition to identify the yield pressure and observe the residual plastic strains and stresses after unloading.

3. Results and Discussion of Task 1

In Task 1, a static structural analysis was performed to evaluate the stress distribution and safety factors of the pressure hull under hydrostatic pressures corresponding to depths of 2000 m, 3800 m, and 4500 m. The results reveal critical insights into the structural behavior of the composite cylinder and titanium endcaps.

3.1. Stress Distribution Analysis

The Von-Mises stress distribution across the whole body indicates that the stress magnitude increases linearly with depth, consistent with the linear nature of hydrostatic pressure $P = \rho g h$. [1]

- **Cylinder Behavior:** The cylindrical section, made of Carbon Fiber Composite, experiences significant compressive stresses. As expected from pressure vessel theory, the Hoop Stress (σ_{yy}) is the dominant stress component and is observed to be approximately twice the magnitude of the Axial Stress (σ_{xx}). This aligns with the theoretical prediction that hoop stress is the primary failure driver in cylindrical vessels. Please examine the figures 5, 6, 11, 12, 17, 18.
- **Hemispherical Cap Behavior:** The Titanium endcaps generally exhibit lower stress levels compared to the cylinder due to their spherical geometry, which distributes the external pressure more efficiently ($\sigma_{sphere} \approx \sigma_{cylinder}/2$). Please see the figures 7,13 and 19.

3.2. Safety Factor Assessment

The safety factors were computed based on the Yield Strength of the materials (600 MPa for Carbon Fiber and 880 MPa for Ti-6Al-4V).

The analysis identifies the Cylinder as the critical component with the lowest safety factor in all cases. For instance, at the design depth of 3800 m, the minimum safety factor for the cylinder was found to be 2.507 whereas the cap maintained a higher safety factor of 3.74 from the figures 14 and 15.

As depth increases to 4500 m, the safety factor decreases further, approaching the design limits. This highlights the necessity of using high-strength composite materials for deep-sea exploration. All the safety factors for different pressure levels can be found in figures 8,9,14,15,20 and 21.

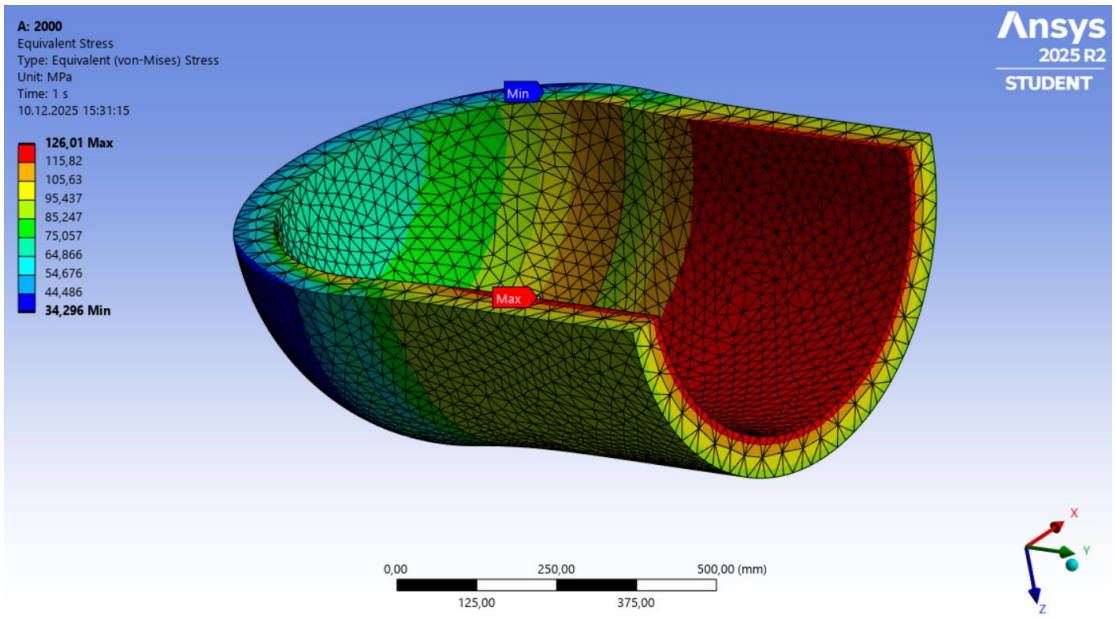


Figure 4: Equivalent (Von-Mises) Stress distribution on the Whole Body at 2000 m depth.

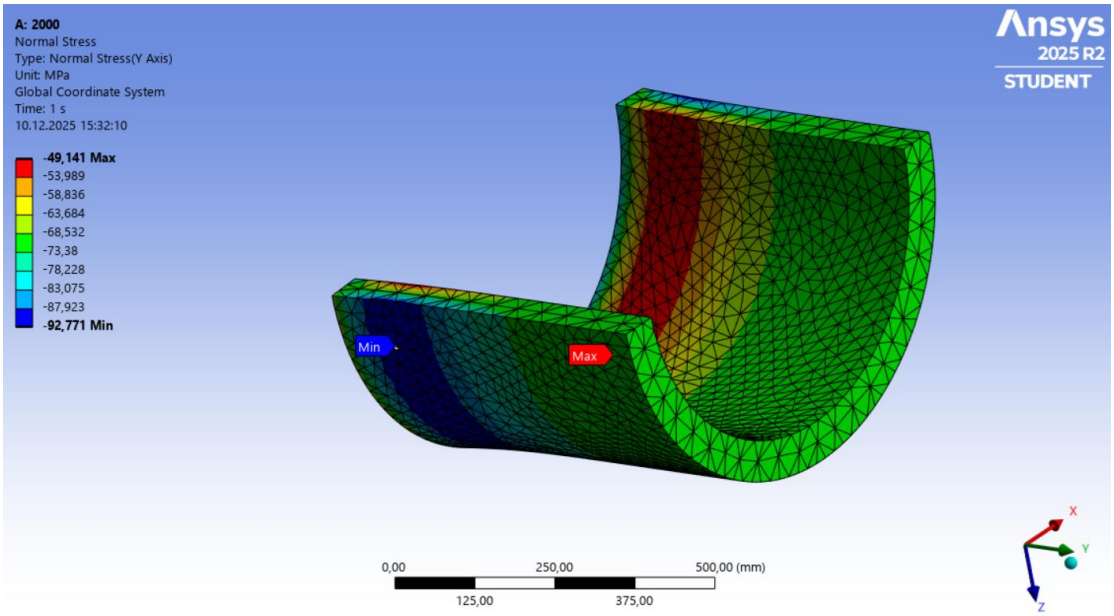


Figure 5: Axial Stress (σ_{xx}) distribution on the Cylinder at 2000 m depth.

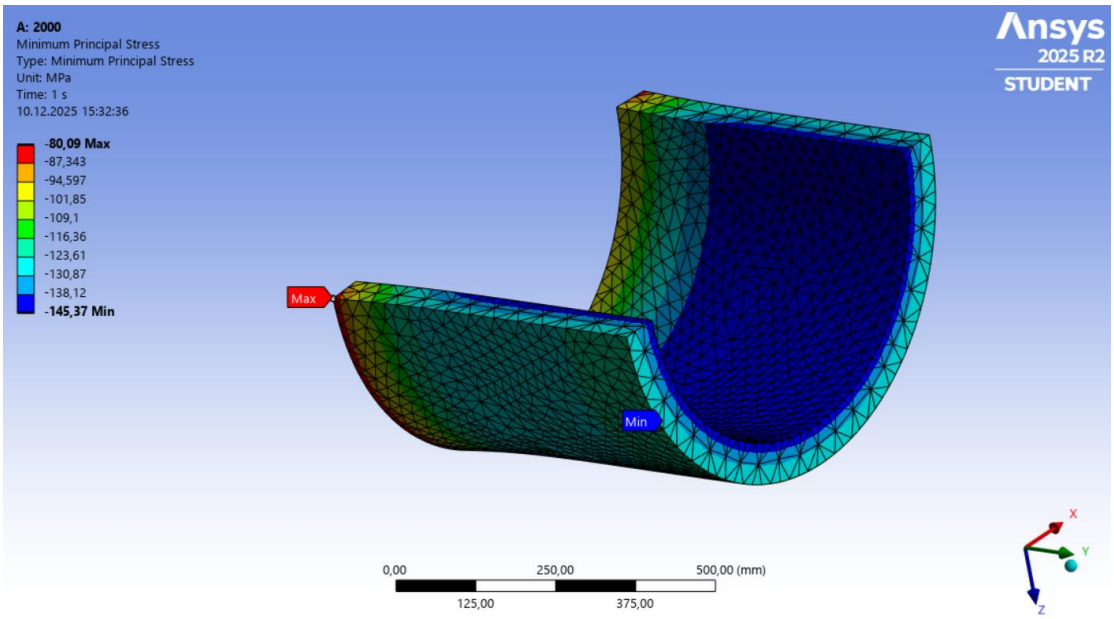


Figure 6: Hoop Stress (σ_{yy}) distribution on the Cylinder at 2000 m depth.

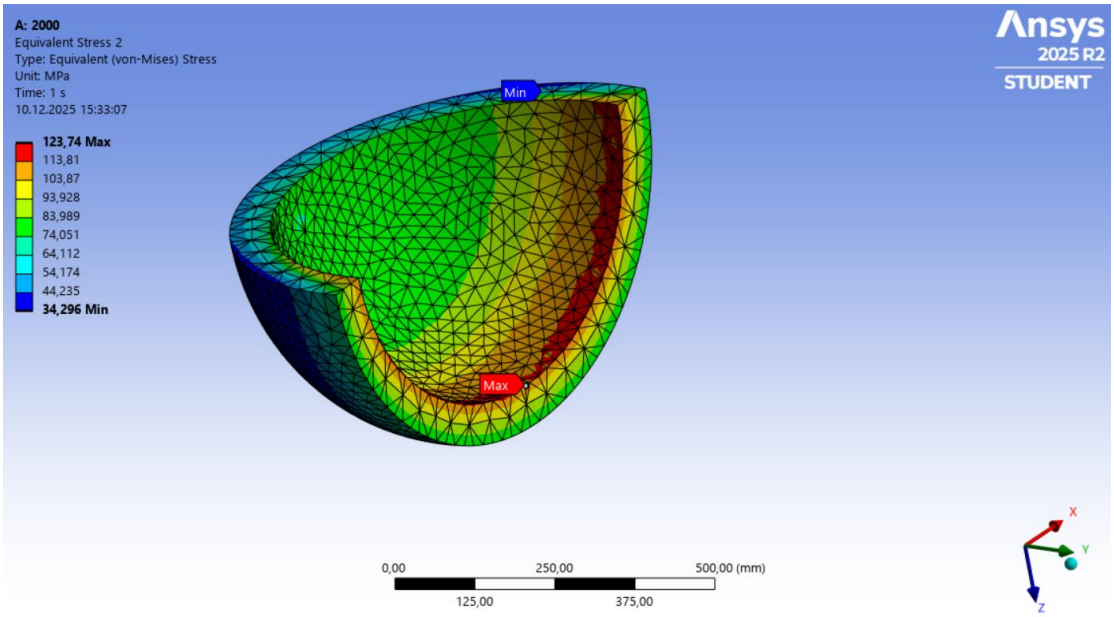


Figure 7: Equivalent (Von-Mises) Stress distribution on the Hemispherical Cap at 2000 m depth.

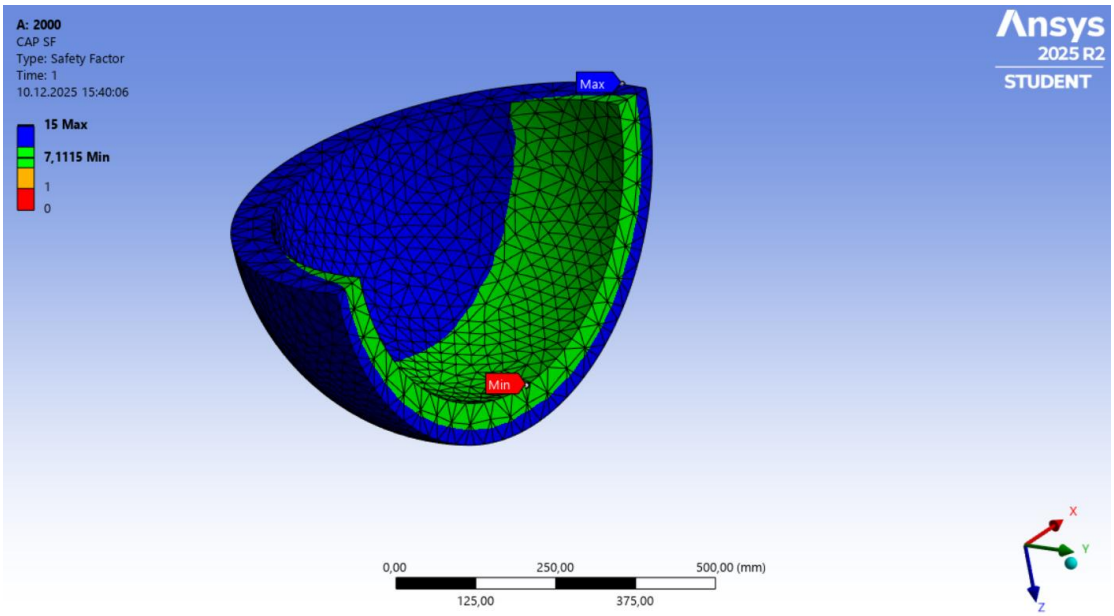


Figure 8: Safety Factor distribution on the Cylinder at 2000 m depth.

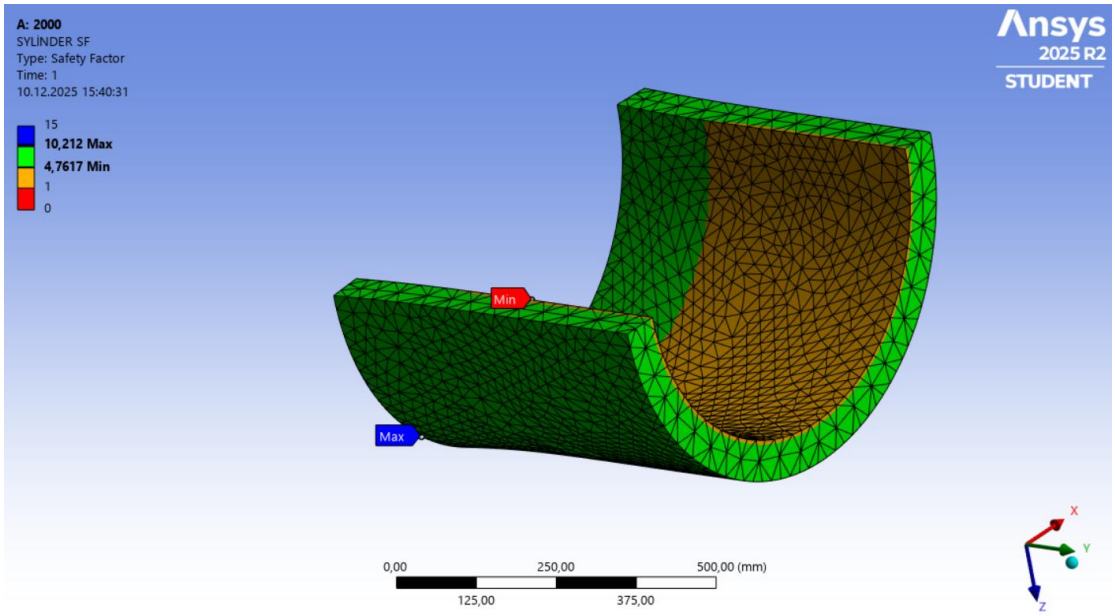


Figure 9: Safety Factor distribution on the Hemispherical Cap at 2000 m depth.

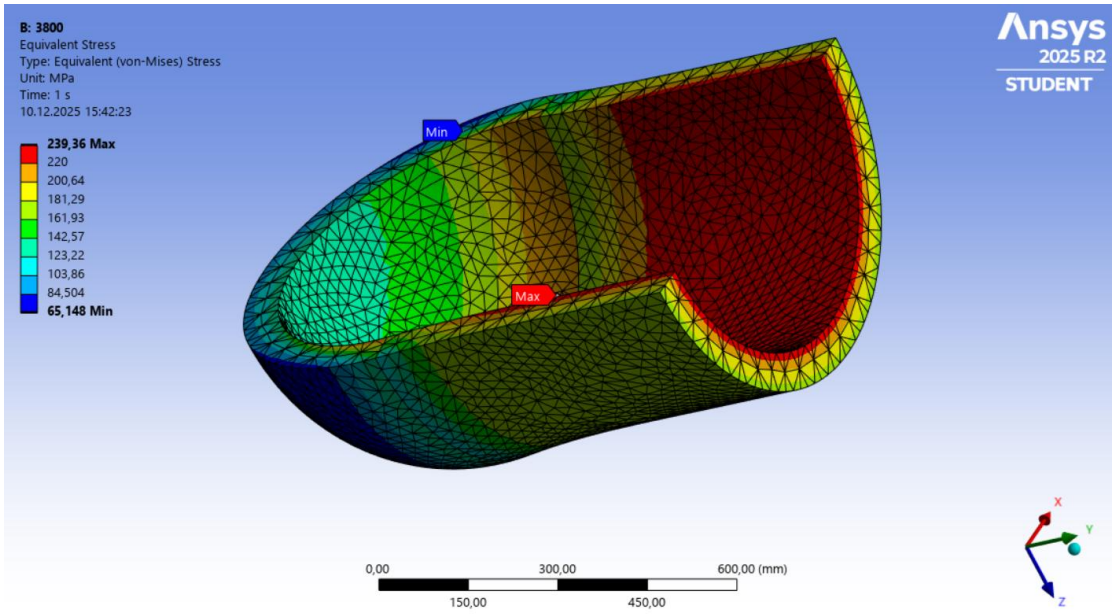


Figure 10: Equivalent (Von-Mises) Stress distribution on the Whole Body at 3800 m depth.

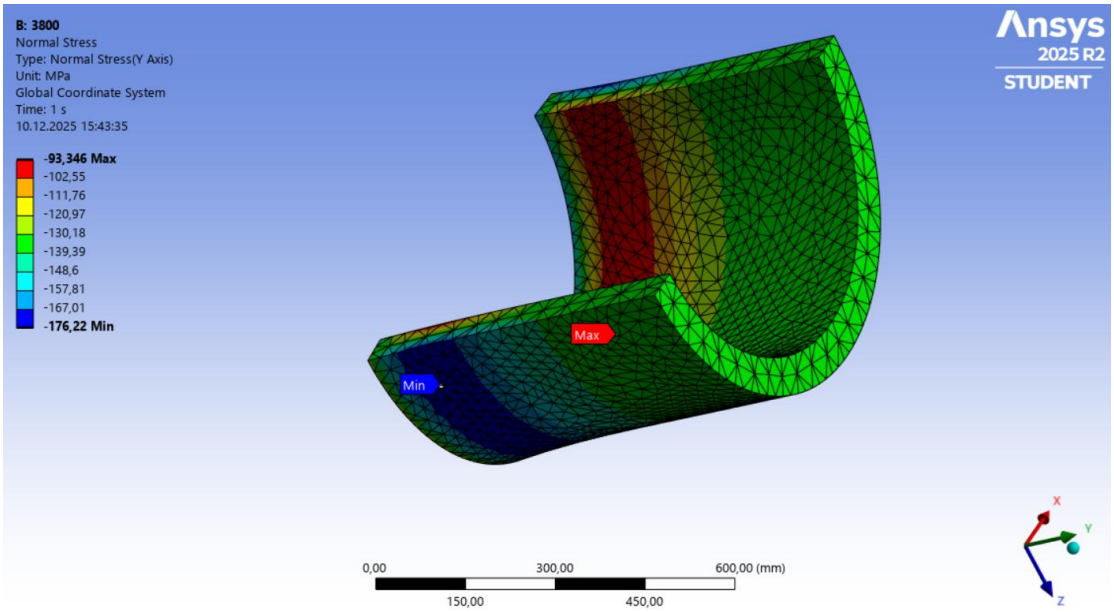


Figure 11: Axial Stress (σ_{xx}) distribution on the Cylinder at 3800 m depth.

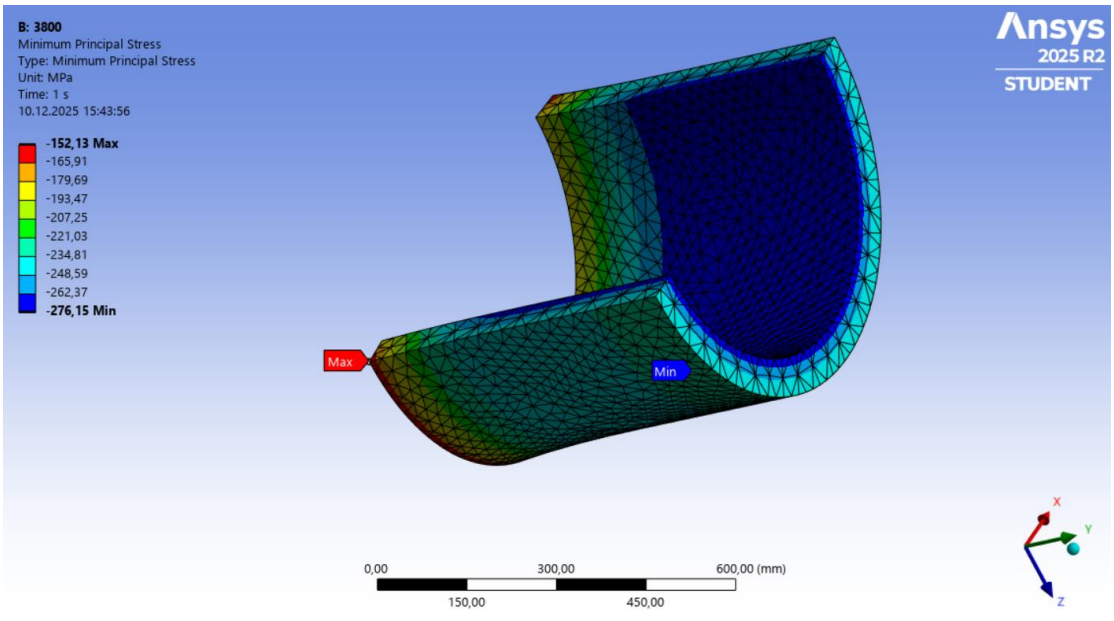


Figure 12: Hoop Stress (σ_{yy}) distribution on the **Cylinder** at 3800 m depth.

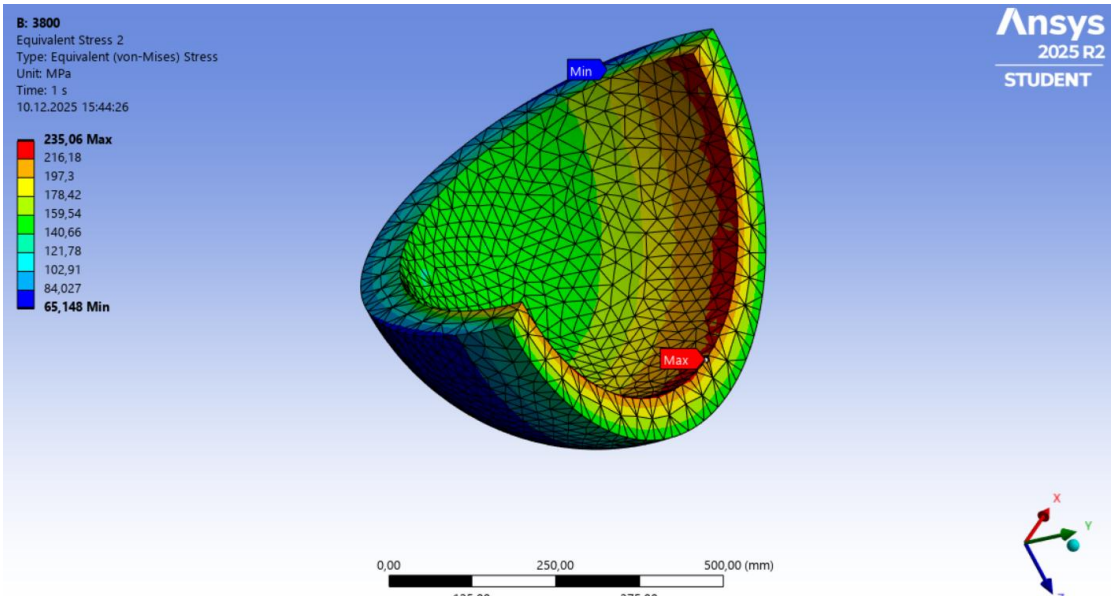


Figure 13: Equivalent (Von-Mises) Stress distribution on the Hemispherical Cap at 3800 m depth.

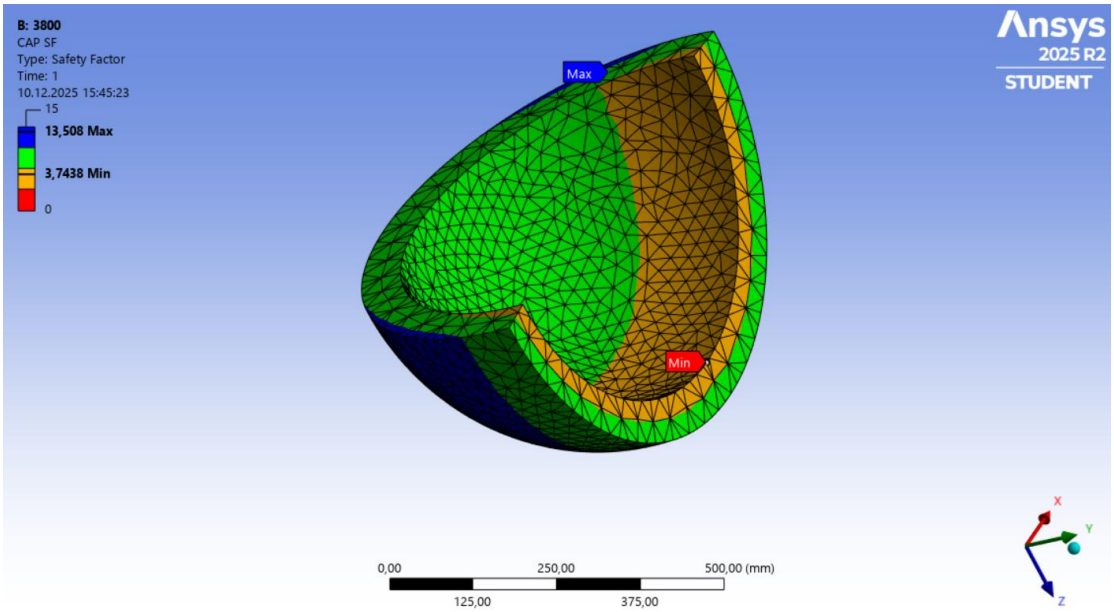


Figure 14: Safety Factor distribution on the Hemispherical Cap at 3800 m depth.

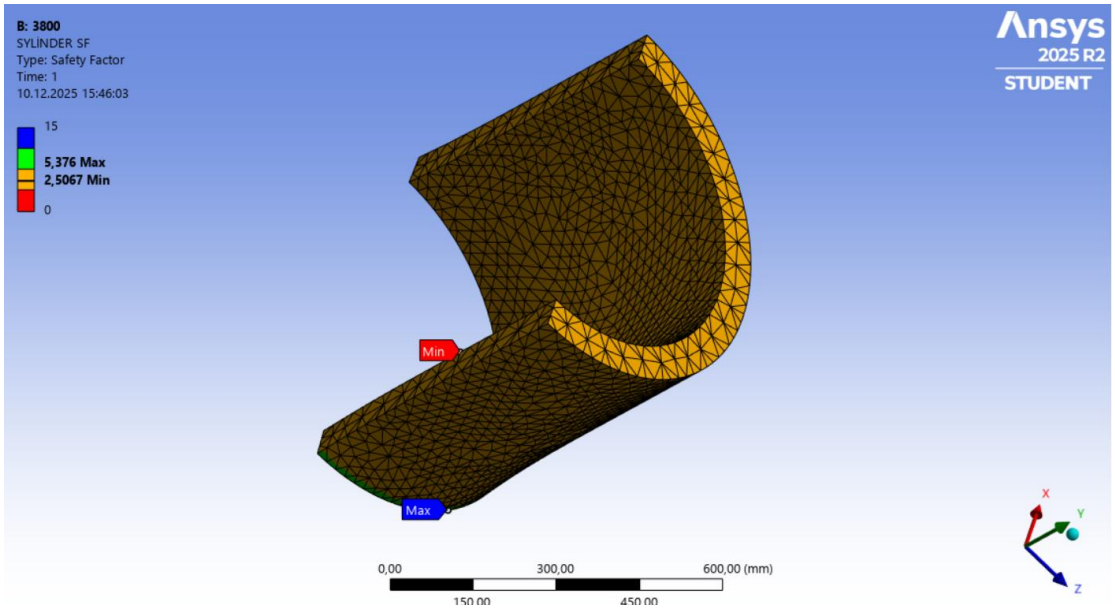


Figure 14: Safety Factor distribution on the Cylinder at 3800 m depth.

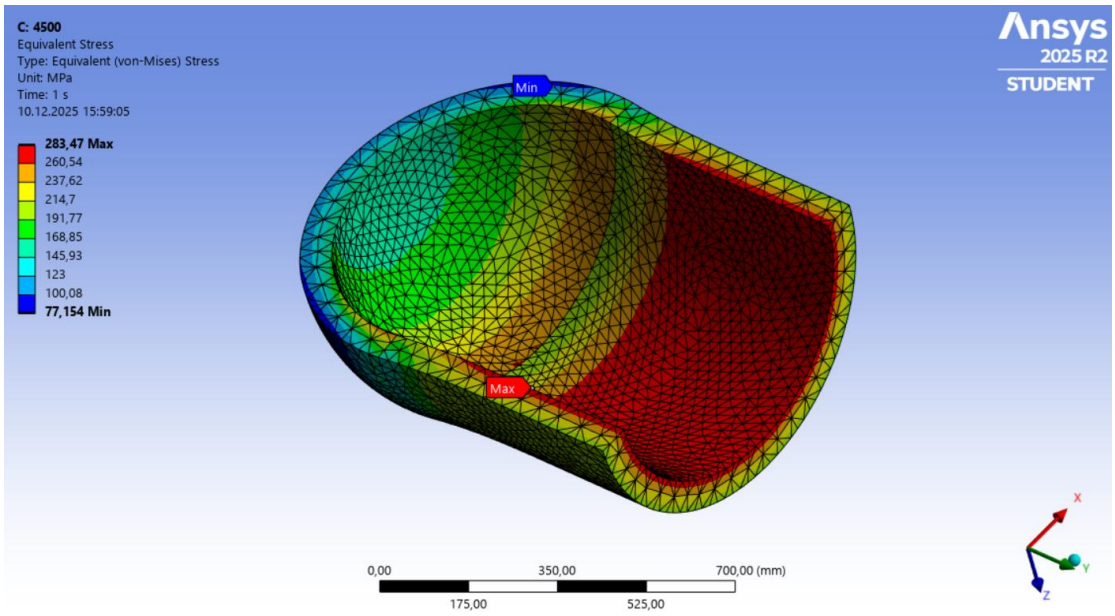


Figure 16: Equivalent (Von-Mises) Stress distribution on the Whole Body at 4500 m depth.

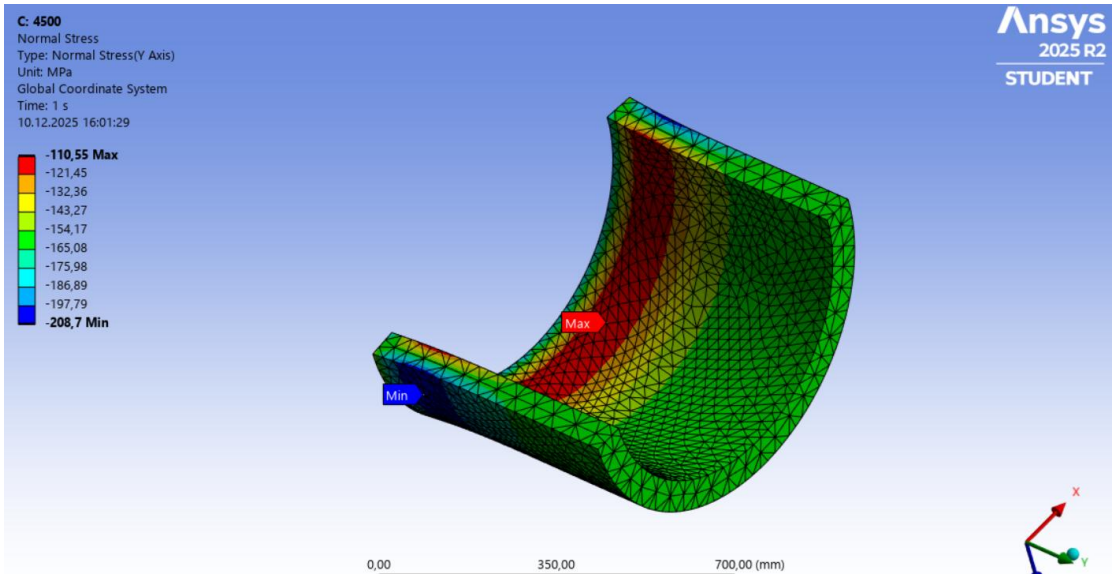


Figure 17: Axial Stress (σ_{xx}) distribution on the **Cylinder** at 4500 m depth.

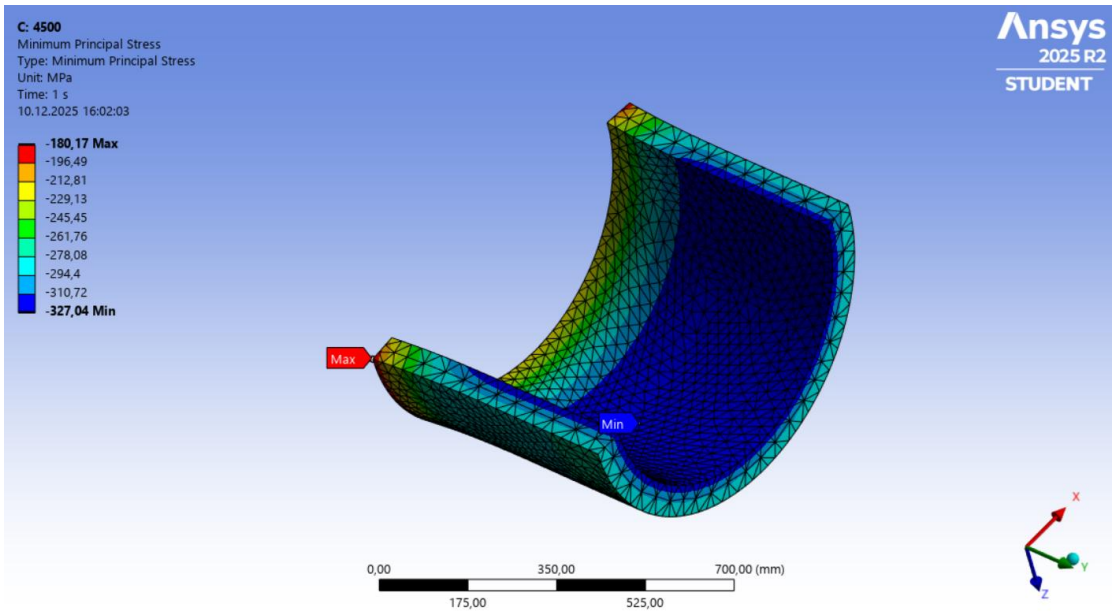


Figure 18: Hoop Stress (σ_{yy}) distribution on the **Cylinder** at 4500 m depth.

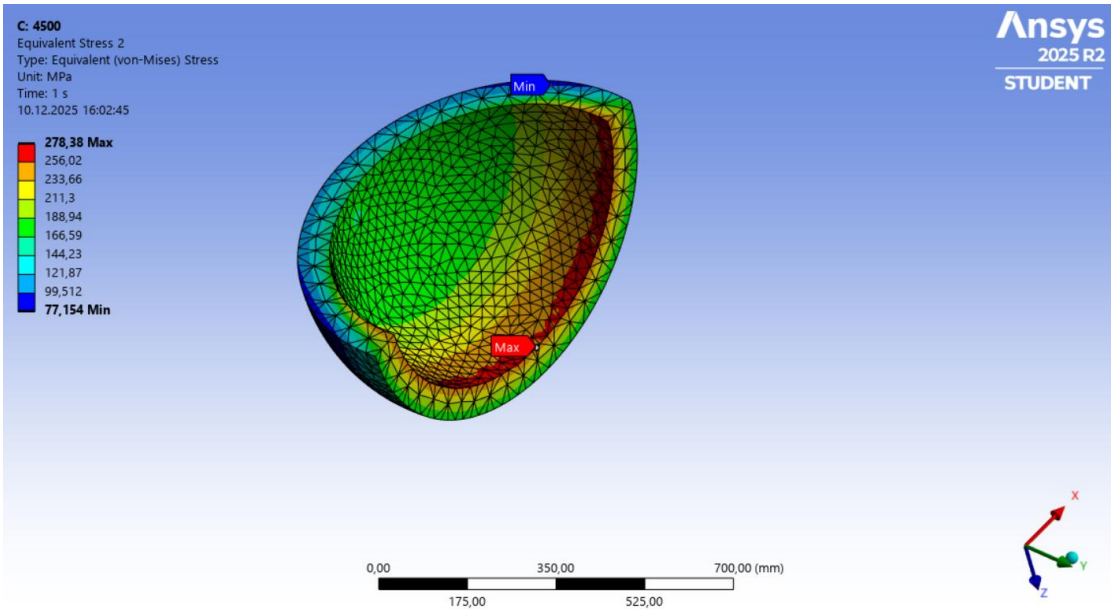


Figure 19: Equivalent (Von-Mises) Stress distribution on the Hemispherical Cap, at 4500 m depth.

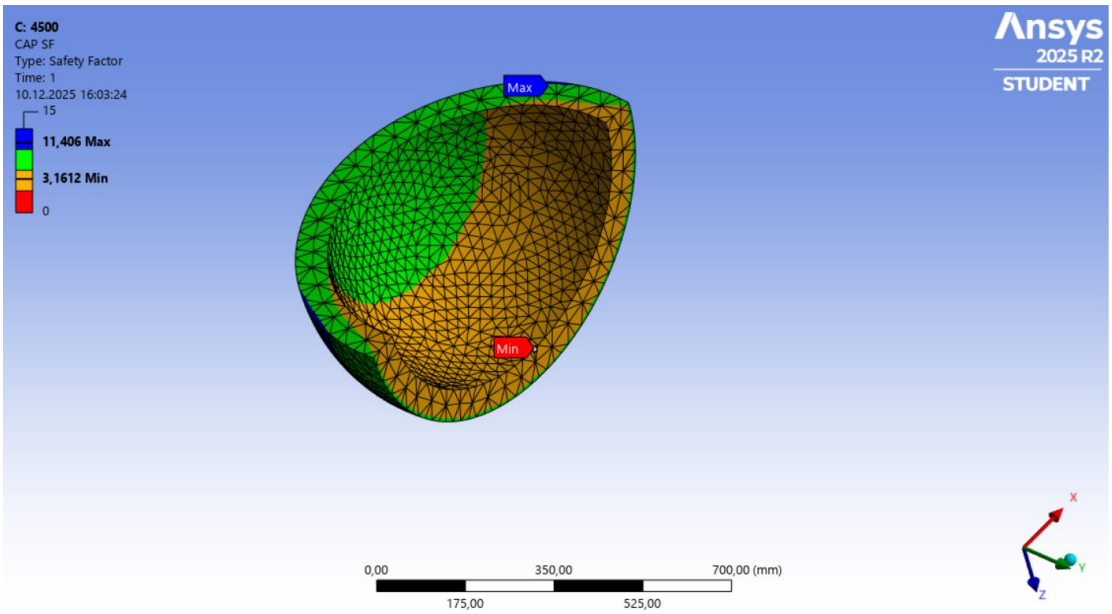


Figure 20: Safety Factor distribution on the Hemispherical Cap at 4500 m depth.

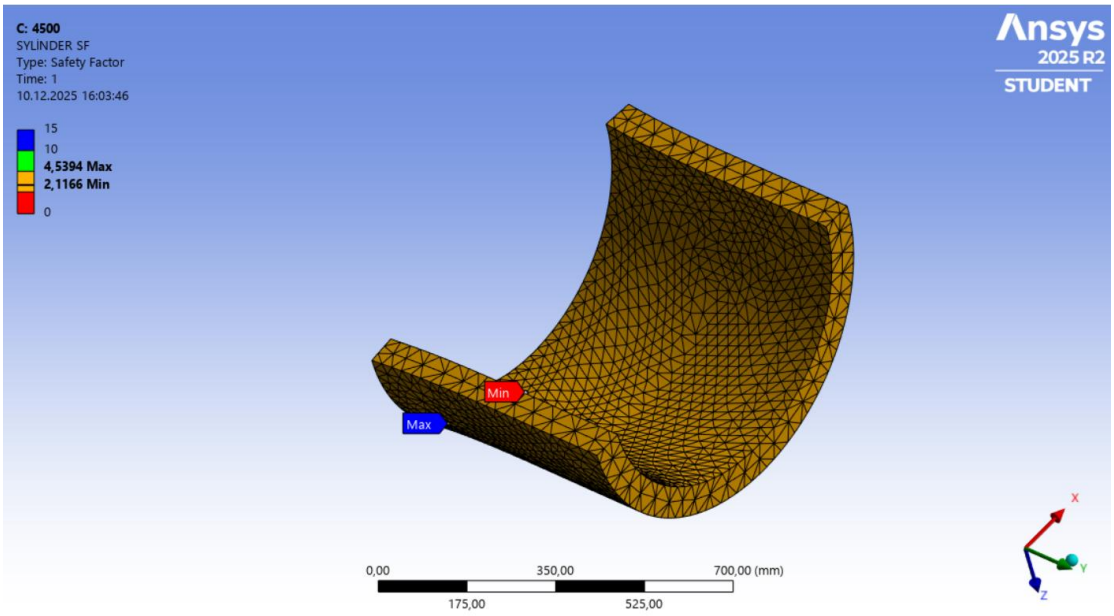


Figure 21: Safety Factor distribution on the Cylinder at 4500 m depth.

3.3. Verification and Validation of Results

To ensure the accuracy of the Finite Element Analysis (FEA), the obtained results were validated against theoretical analytical solutions referenced from Young, W.C., "Roark's Formulas for Stress and Strain", McGraw-Hill.

Since the ratio of the outer radius to thickness $(R_o/t \approx 418.5/63.5 \approx 6.6)$ is relatively low, the vessel acts as a thick-walled structure. However, thin-walled approximations provide a useful first-order verification.

Check 1 - Stress Magnitude:

According to the thin-walled formula for a cylinder, the Hoop Stress is

$$\sigma_h = \frac{PR}{t}$$

At 3800 m ($P \approx 38.2$ MPa), the theoretical hoop stress is approximately:

$$\sigma_h \approx \frac{38.2 \times 355}{63.5} \approx 213 \text{ MPa}$$

The FEA result for Hoop Stress (Minimum Principal Stress) at the mid-span of the cylinder is 221MPa from Figure 12. The close agreement between the theoretical value and the FEA result confirms that the mesh quality and boundary conditions are correctly defined.

Check 2 - Geometric Relationship:

Theoretical mechanics dictate that for the same thickness and radius, the stress in a spherical cap should be approximately half of the hoop stress in a cylinder

$$\sigma_{sphere} = \frac{PR}{2t}$$

.

Our ANSYS results show that the average stress in the Cap is indeed lower than the Cylinder Hoop stress, validating the physical behavior of the model. From figures 4,10 and 16 it can be inferred that average stresses are more on Cylinder.

Check 3 - Boundary Effects:

The symmetry boundary conditions produced a realistic deformation pattern with zero axial displacement at the cut faces, ensuring that the model correctly simulates a continuous closed vessel without rigid body motion.

4. Results and Discussion of Task 2: Parametric Study

In Task 2, a parametric study was conducted to investigate the influence of wall thickness on the structural safety of the pressure hull. The depth was kept constant at the design depth of 3800 m (approx. 38.2 MPa), while the wall thickness (T_c and T_h) was varied across four configurations: 40 mm, 60 mm, 80 mm, and 100 mm.

4.1. Effect of Thickness on Safety Factor

The Minimum Safety Factors (SF) obtained for both the Carbon Fiber Cylinder and the Titanium Cap are summarized in Table 2 and visualized in Figure 22.

Thickness (mm)	40	60	80	100
Cylinder Min. Safety Factor	1.65	2.4	3.12	3.8
Cap Min. Safety Factor	2.57	3.82	4.95	5.99

Table 2: The effect of wall thickness (t) on the safety factor was examined at

3800 m depth

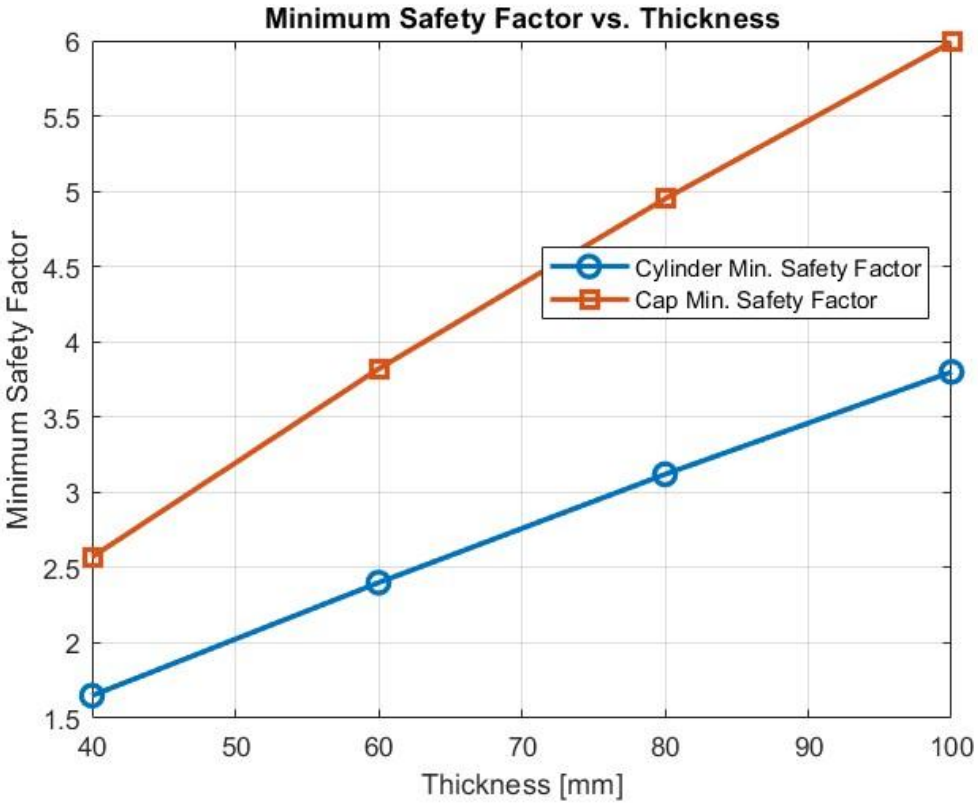


Figure 22: Graph showing the relationship between Minimum Safety Factor and Wall Thickness (mm) for both Cylinder and Cap at 3800 m depth.

The results demonstrate a clear positive linear correlation between wall thickness and the safety factor. As the thickness increases from 40 mm to 100 mm, the safety factor of the cylinder improves significantly from 1.65 to 3.8. Similarly, the cap's safety factor increases from 2.57 to 5.99.

4.2. Physical Interpretation of the Trend

The observed trend can be explained by the fundamental mechanics of pressure vessels. The membrane stress (σ) in a thin-walled vessel is inversely proportional to the thickness (t), as governed by the relationship

$$\sigma \propto \frac{PR}{t}$$

Therefore, increasing the wall thickness increases the cross-sectional area available to resist the external hydrostatic pressure. This reduction in effective stress, while the Material Yield Strength remains constant, leads to a direct increase in the Safety Factor

$$SF = \frac{\sigma_{yield}}{\sigma_{max}}$$

Consistent with Task 1, the Cylinder remains the critical component (weakest link) in all thickness configurations, exhibiting lower safety factors than the hemispherical caps due to the higher hoop stresses inherent to cylindrical geometries.

5. Results and Discussion of Task 3: Elasto-Plastic Analysis

In Task 3, the analysis was extended beyond linear elasticity to investigate the post-yield behavior of the Titanium (Ti-6Al-4V) hemispherical caps. A Bilinear Isotropic Hardening material model was applied, defined by a Yield Strength of 880 MPa and a Tangent Modulus of 2000 MPa. The simulation utilized a two-step "Load-Unload" sequence to distinguish between recoverable elastic strain and permanent plastic deformation.

The study was conducted under three distinct pressure levels: Design Pressure (~38.2 MPa), Estimated Yield Pressure (~152 MPa), and 50% Above Yield Pressure (~228 MPa).

5.1. Elastic Response at Design Pressure (38.2 MPa)

At the design depth of 3800 m, the pressure hull operates well within the safe limits. The results show that the Equivalent Von-Mises stress remains significantly below the material's yield strength of 880 MPa. Upon removing the load (at Time = 2.0 s), both the Von-Mises stress and the Equivalent Elastic Strain return completely to zero. As expected, the Equivalent Plastic Strain is zero throughout the entire process, confirming that the vessel behaves in a perfectly elastic manner under standard operating conditions. Figures 23 to 28.

5.2. Transition at Estimated Yield Pressure (~152 MPa)

Through iterative analysis, the pressure required to initiate yielding in the hemispherical cap was estimated to be approximately 152 MPa. At this pressure

level, the local Von-Mises stress in critical regions (likely near the cylinder-cap junction) reaches the material's Yield Strength (880 MPa). After unloading, a minute but non-zero Equivalent Plastic Strain ($\approx 5 \times 10^{-6}$) was observed. This signifies the exact transition point where the material begins to suffer permanent micro-structural changes. While this deformation is negligible for immediate structural failure, it marks the limit of the vessel's reusability without damage. See Figures 29 to 34.

5.3. Plastic Deformation at 50% Above Yield (~228 MPa)

The most critical insights were observed when the vessel was subjected to an extreme overload of 228 MPa (50% above the yield pressure). The Equivalent Plastic Strain contour plot at Time = 2.0 s (after unloading) reveals significant red zones. Unlike elastic strain, plastic strain does not recover when the load is removed. This confirms that the hemispherical cap has undergone permanent geometric distortion. See Figures 35 and 37.

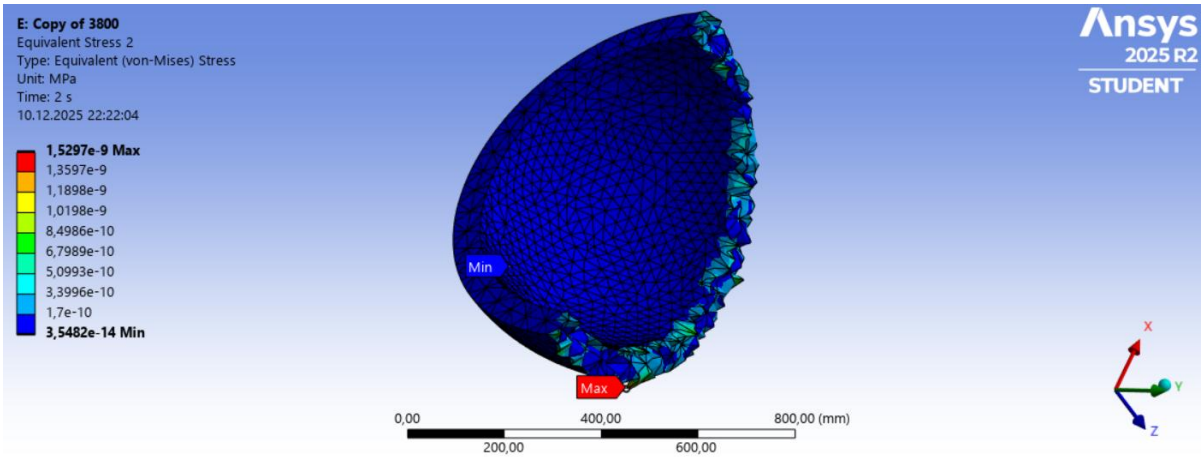


Figure 23: Equivalent (Von-Mises) Stress on the Hemispherical Cap after unloading from 3800 m equivalent pressure.

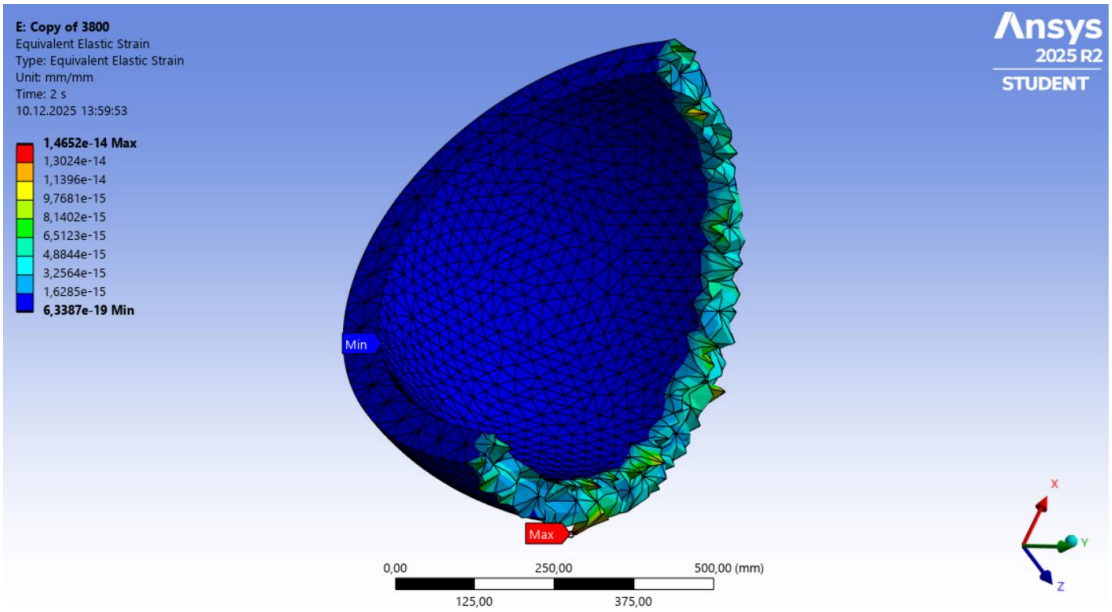


Figure 24: Equivalent Elastic Strain on the Hemispherical Cap after unloading from 3800 m equivalent pressure.

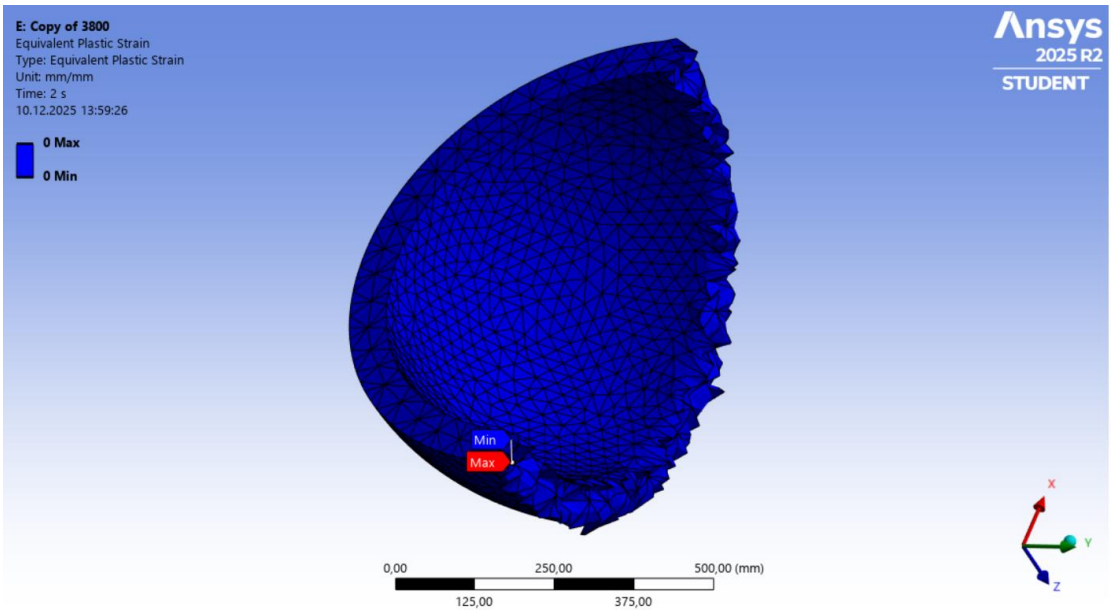


Figure 25: Equivalent Plastic Strain on the Hemispherical Cap after unloading from 3800 m equivalent pressure.

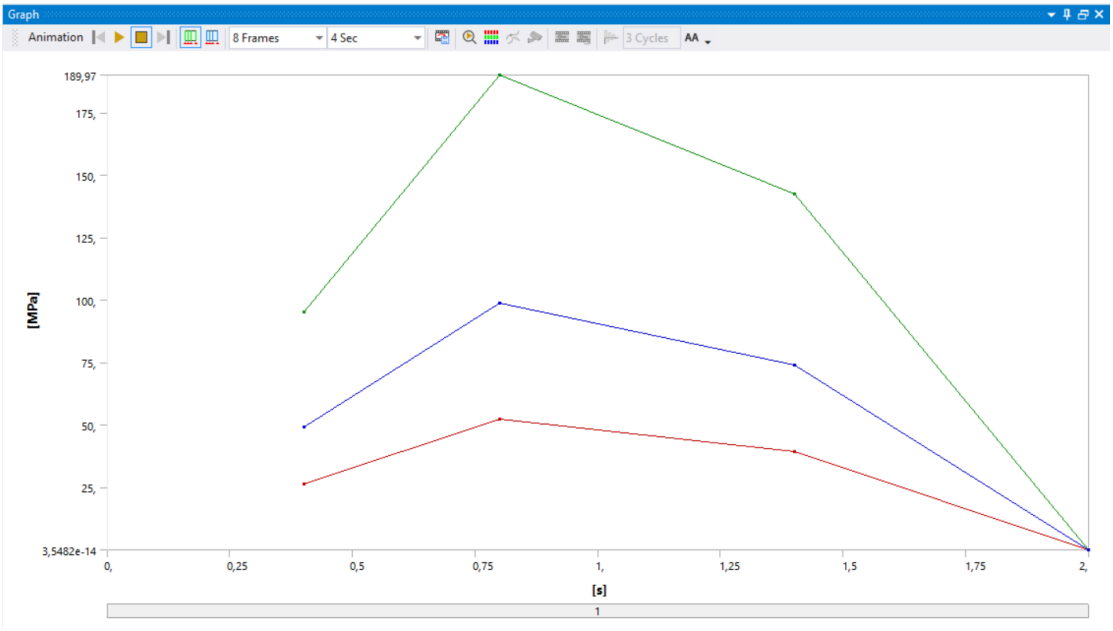


Figure 26: Time-history chart of Von-Mises Stress at 3800 m equivalent pressure load case.

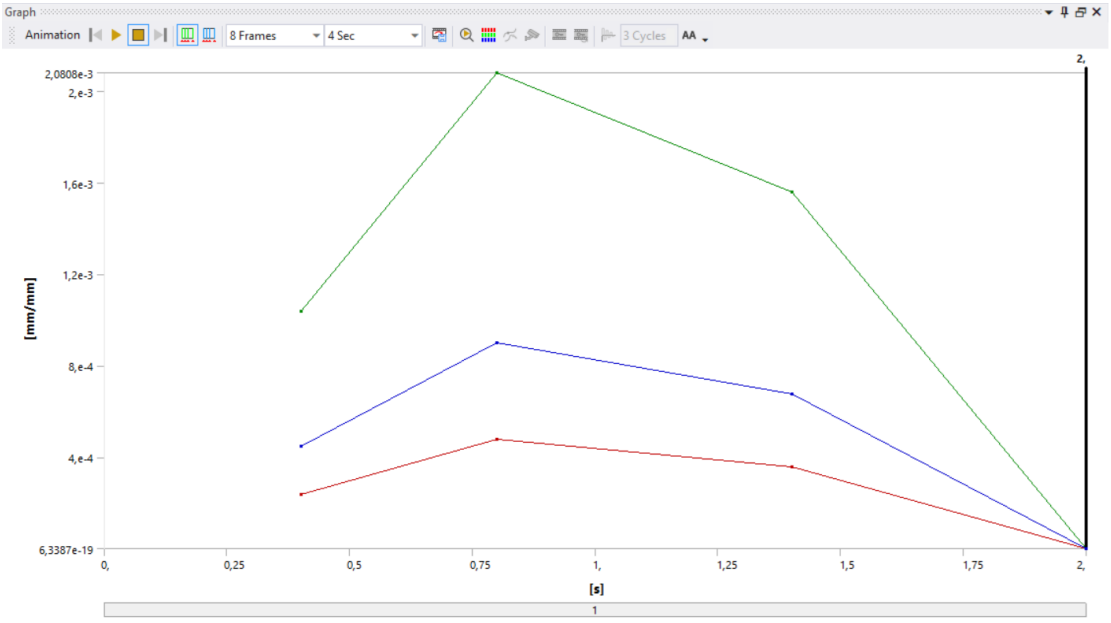


Figure 27: Time-history chart of Elastic Strain at 3800 m equivalent pressure load case.

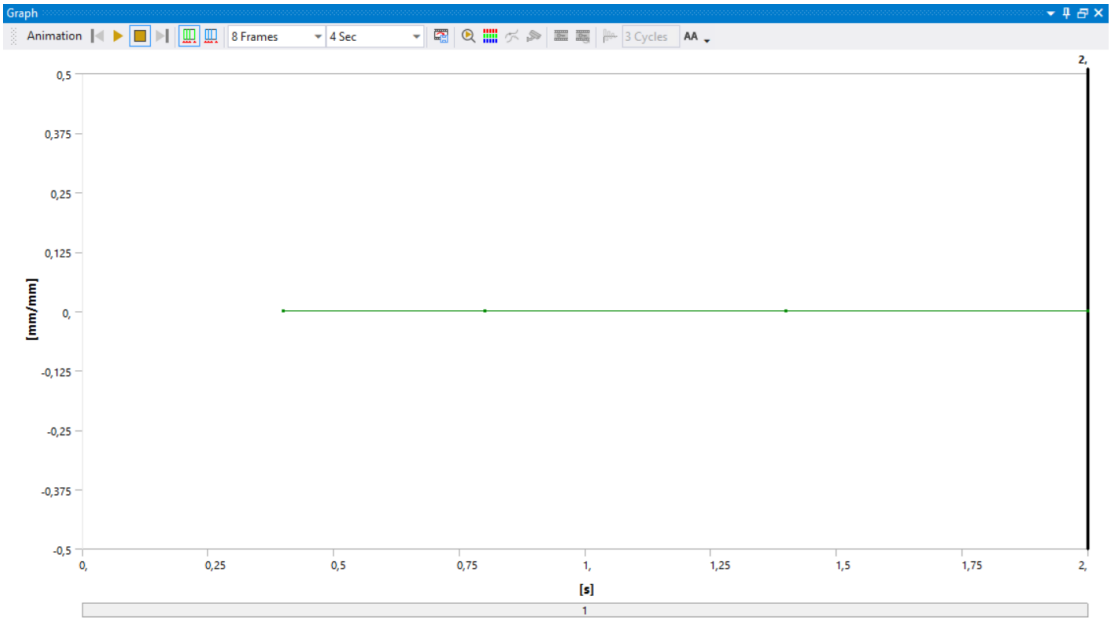


Figure 28: Time-history chart of Plastic Strain at 3800 m equivalent pressure load case.

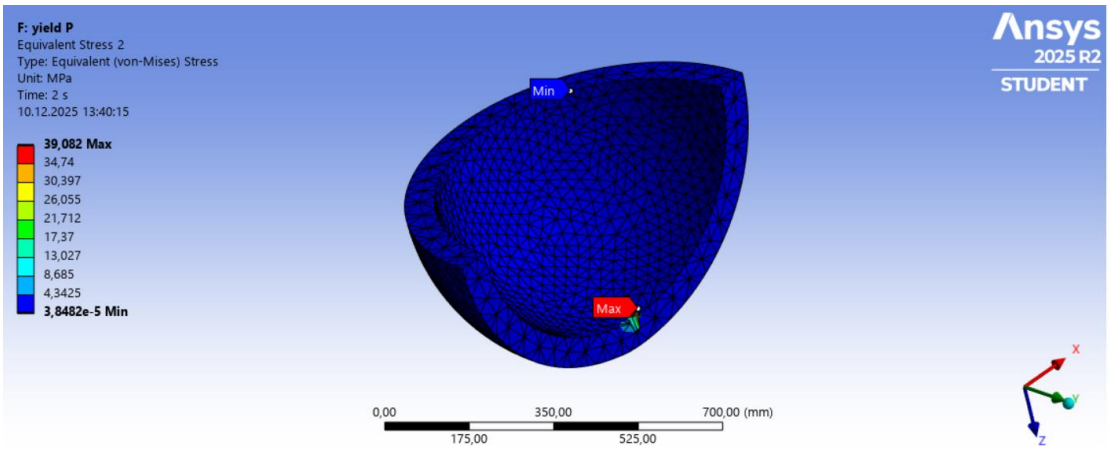


Figure 29: Equivalent (Von-Mises) Stress on the Hemispherical Cap after unloading from Yield Pressure (~152 MPa).

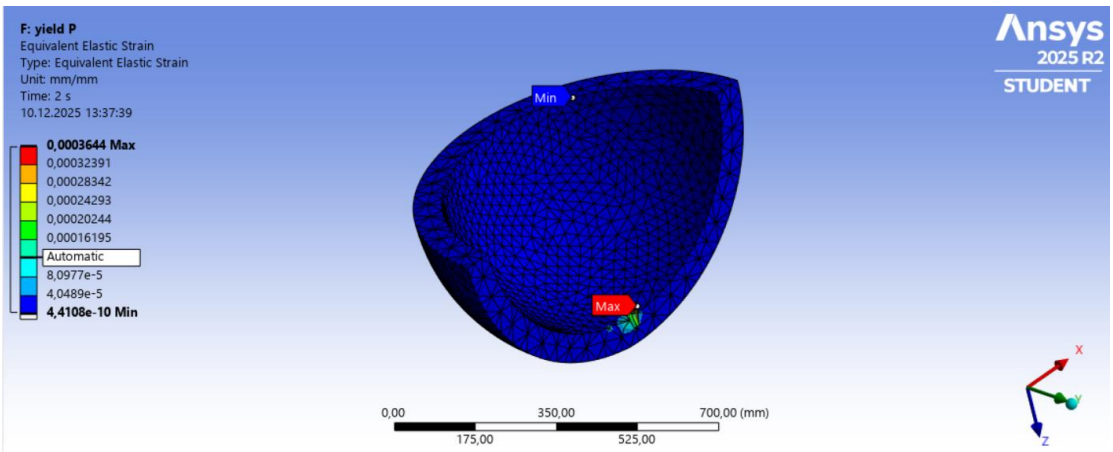


Figure 30: Equivalent Elastic Strain on the Hemispherical Cap after unloading from Yield Pressure (~152 MPa).

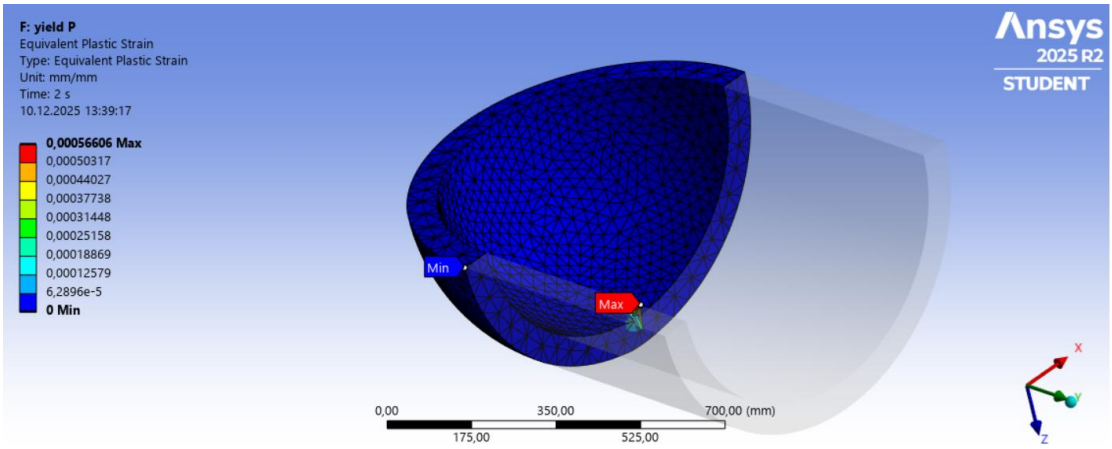


Figure 31: Equivalent Plastic Strain on the Hemispherical Cap after unloading from Yield Pressure (~152 MPa).

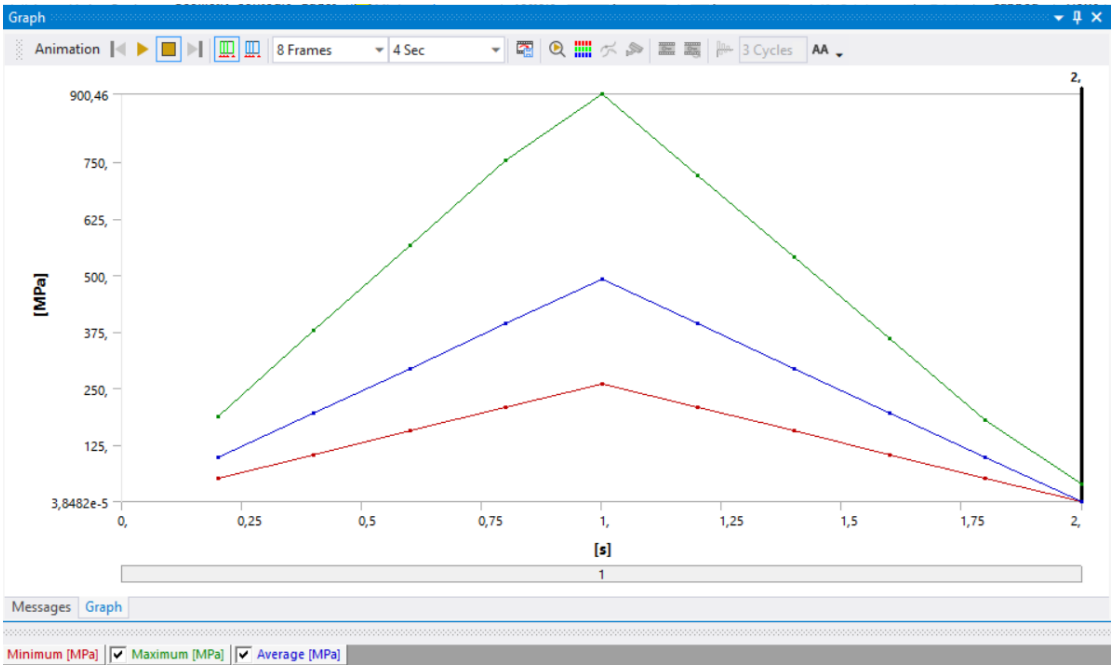


Figure 32: Time-history chart of Von-Mises Stress at Yield Pressure (~152 MPa) load case.

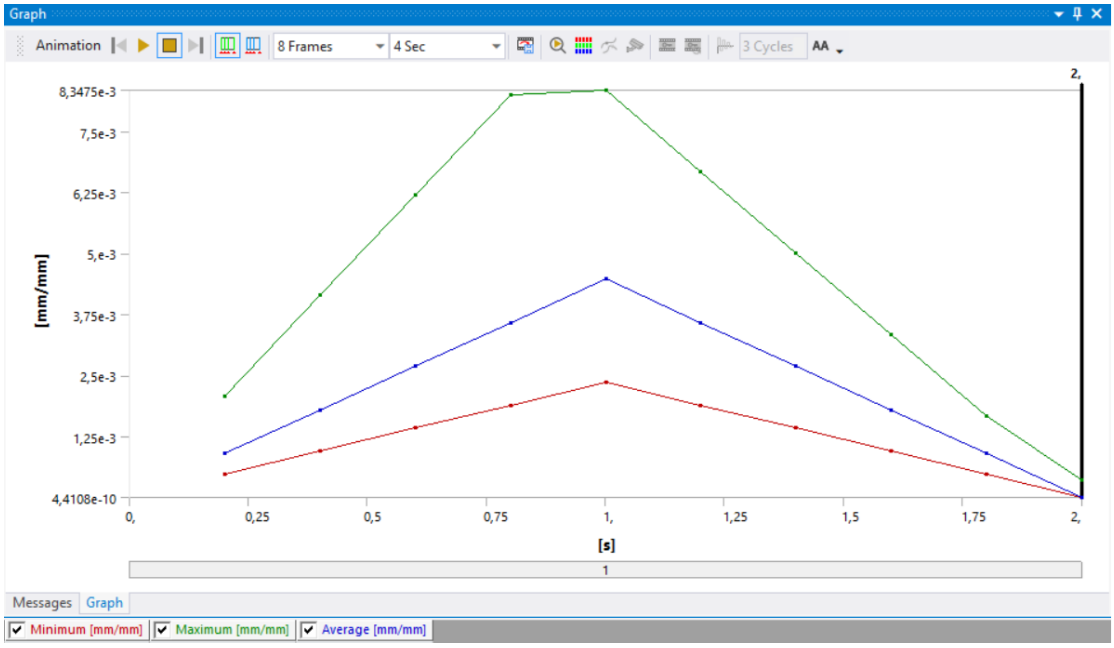


Figure 33: Time-history chart of Elastic Strain at Yield Pressure (~152 MPa) load case.

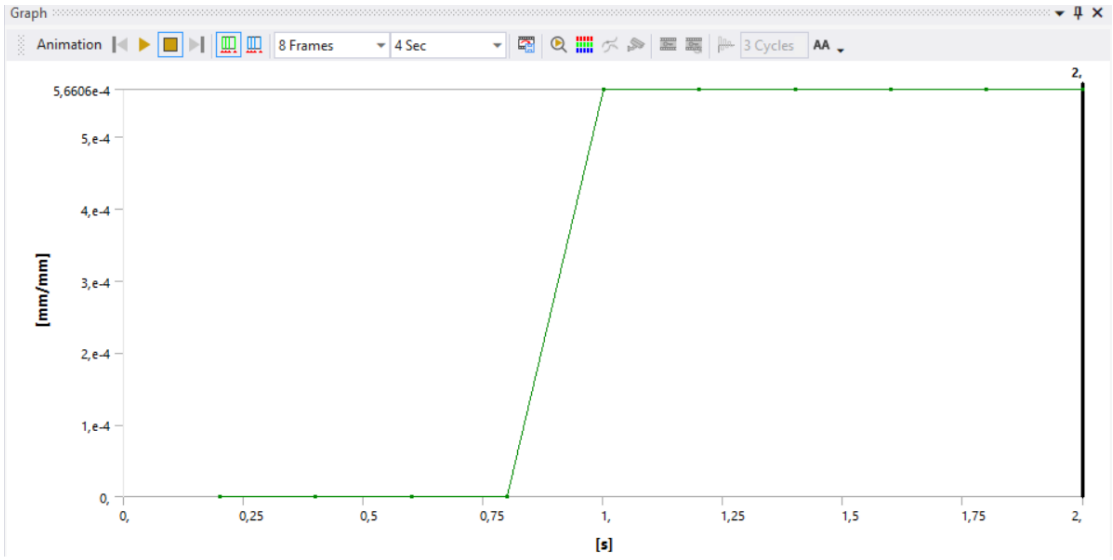


Figure 34: Time-history chart of Plastic Strain at Yield Pressure (~152 MPa) load case.

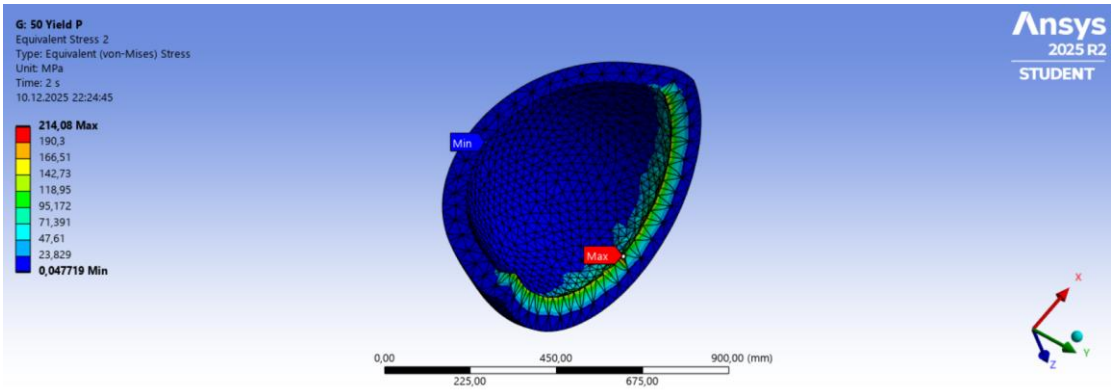


Figure 35: Equivalent (Von-Mises) Stress on the Hemispherical Cap after unloading from 50% above Yield Pressure (~228 MPa).

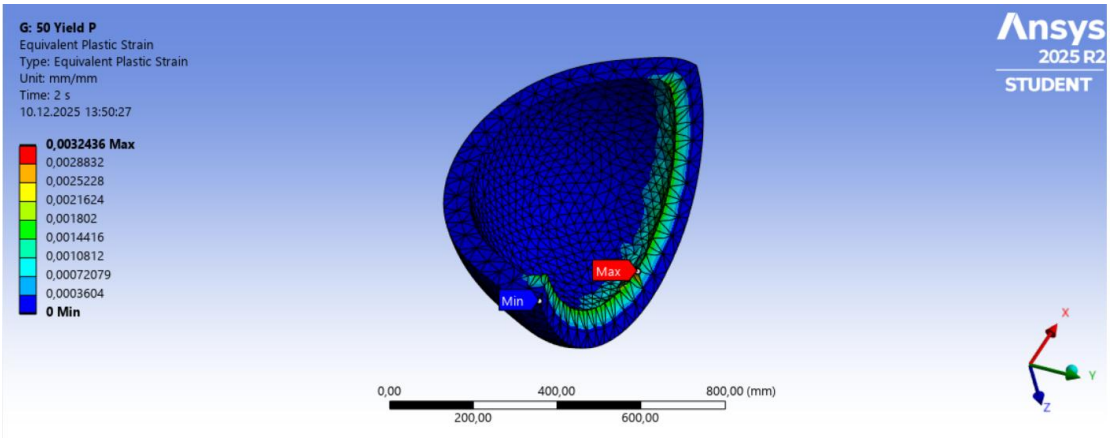


Figure 37: Equivalent Plastic Strain on the Hemispherical Cap after unloading from 50% above Yield Pressure (~228 MPa).

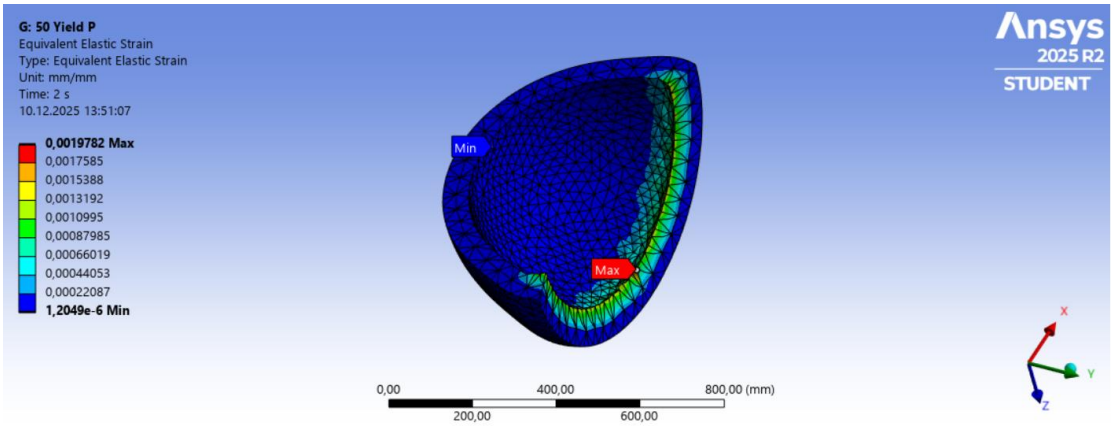


Figure 36: Equivalent Elastic Strain on the Hemispherical Cap after unloading from 50% above Yield Pressure (~228 MPa).

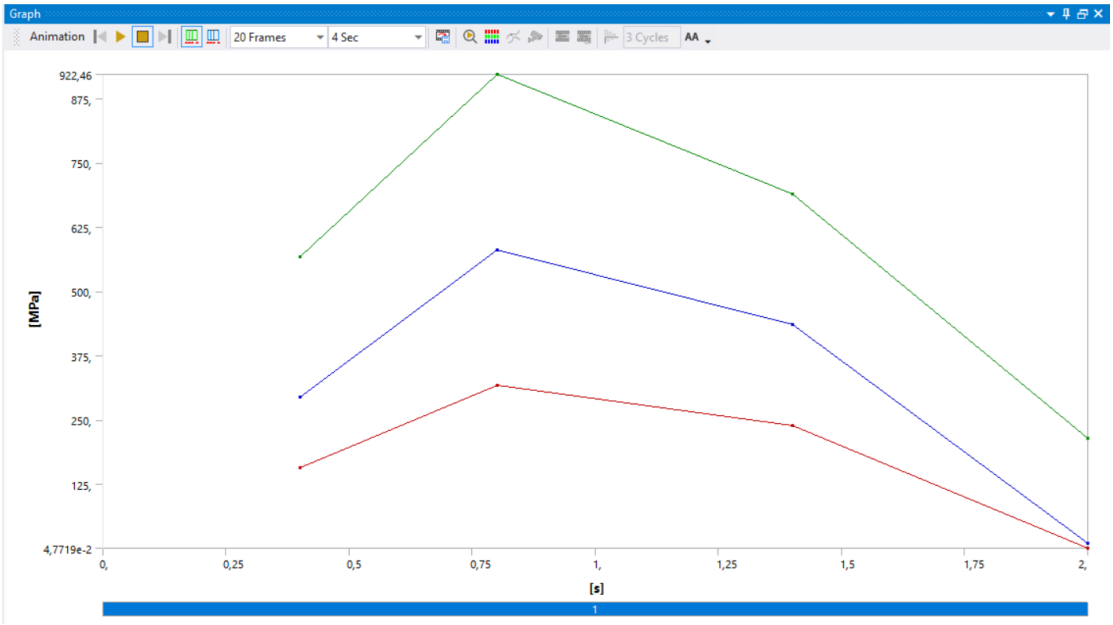


Figure 38: Time-history chart of Von-Mises Stress at 50% above Yield Pressure (~228 MPa) load case.

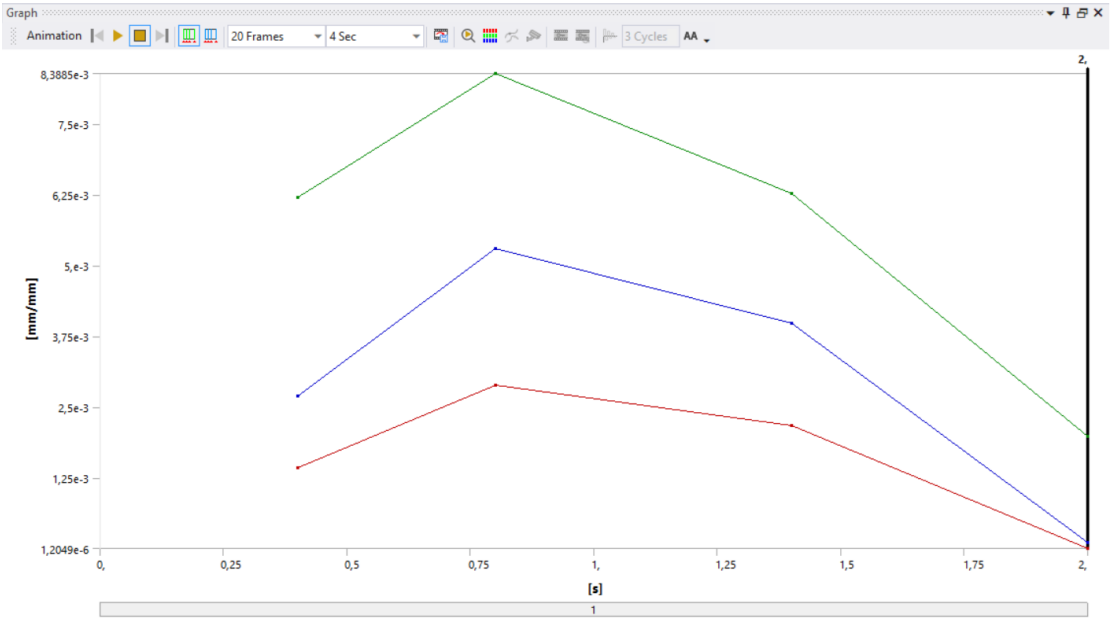


Figure 39: Time-history chart of Elastic Strain at 50% above Yield Pressure (~228 MPa) load case.

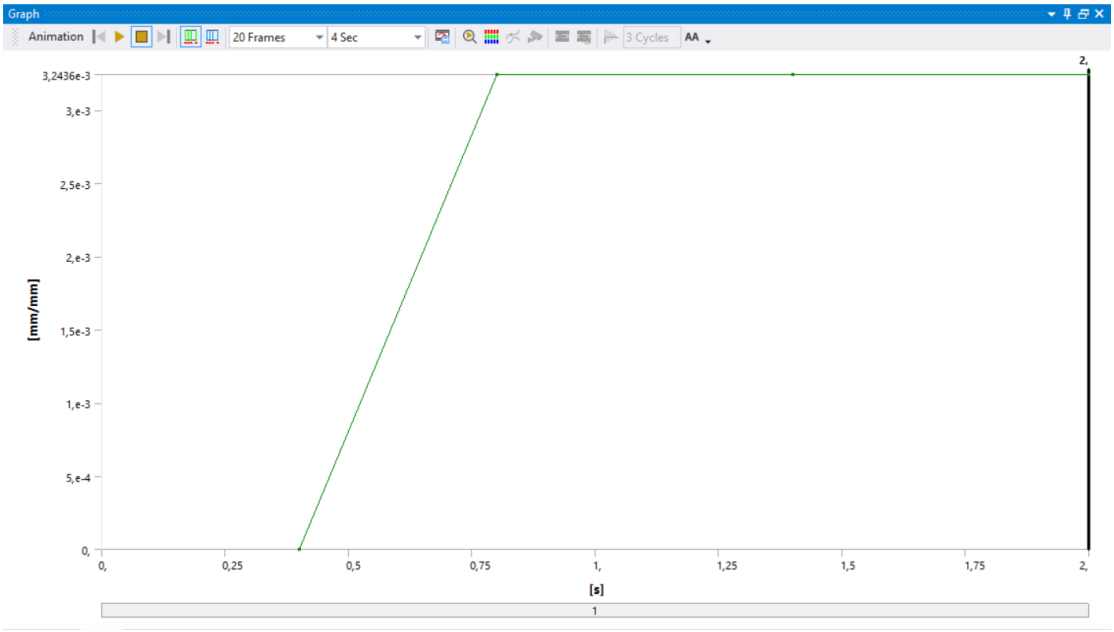


Figure 40: Time-history chart of Plastic Strain at 50% above Yield Pressure (~228 MPa) load case.

6. Discussion and Comparison with Literature & Real-World Context

This section interprets the Finite Element Analysis (FEA) results in the context of theoretical literature, the real-world engineering failure of the OceanGate Titan submersible, and the broader implications for pressure vessel safety in daily engineering applications.

6.1. Validation with Theoretical Literature

The stress results obtained in Task 1 show a high degree of correlation with classical shell theory found in standard textbooks such as Young, W.C., "Roark's Formulas for Stress and Strain". The simulation confirmed that the Hoop Stress in the cylindrical section is approximately twice the Axial Stress ($\sigma_{hoop} \approx 2\sigma_{axial}$). This theoretical behavior is the primary reason why cylindrical pressure vessels are more prone to failure than spherical ones under identical pressure. The FEA results highlighted a stress concentration at the bonded interface between the Carbon Fiber Cylinder (E = 80 GPa) and the Titanium Cap (E=110 GPa). Literature on composite-metal joints confirms that this "stiffness discontinuity" is a critical region where shear stresses develop, leading to delamination or debonding in real-world scenarios.

6.2. Engineering Insights from the OceanGate Titan Incident

Although this project utilizes a simplified isotropic material model and does not simulate the complex buckling instability that likely caused the Titan implosion, the structural stress analysis offers vital clues regarding the catastrophe.The analysis consistently identified the Cylinder as the component with the lowest Safety Factor compared to the hemispherical caps4. This aligns with the forensic understanding of the Titan incident, where the failure originated in the carbon fiber cylindrical hull rather than the titanium endcaps. In Task 3, the Titanium cap showed post-yield "hardening" behavior, meaning it can deform plastically before breaking. In contrast, Carbon Fiber (the cylinder material) is brittle; it remains linear elastic until it shatters catastrophically. The simulation underscores the risk of using brittle materials for the primary pressure boundary, as they offer no warning signs (like bulging) prior to failure. In Task 2, a wall thickness of 40 mm resulted in a Safety Factor of only 1.65. In deep-sea engineering, where human life is at risk, standard classification societies require much higher safety margins to account for manufacturing defects, fatigue, and material inconsistencies. The analysis suggests that operating with low safety margins leaves no room for error.

6.3. The Significance of Residual Strain (Task 3)

The elasto-plastic study in Task 3 demonstrated that once the material exceeds its Yield Pressure (approx. 152 MPa), it retains permanent plastic strain after unloading. This is directly relevant to "Low-Cycle Fatigue." A submersible dives and resurfaces repeatedly (Load-Unload cycles). If a dive inadvertently pushes the hull slightly beyond its elastic limit—or if the yield strength degrades over time the hull accumulates "locked-in" stresses and permanent deformation. The "Plastic Strain vs. Time" charts (Figures 34 and 40) visually prove that this damage is irreversible. In the real world, this accumulated damage from previous dives likely contributed to the eventual fatigue failure of the Titan, emphasizing why rigorous non-destructive testing is required after every dive.

7. Conclusion

In conclusion, the structural analysis of the composite-metal pressure hull confirmed that the Carbon Fiber Cylinder is the critical "weak link" in the assembly, exhibiting lower safety factors than the Titanium caps due to

dominant hoop stresses. The parametric study revealed a significant dependency of safety on geometry, showing that increasing the wall thickness from 40 mm to 100 mm improves the safety factor from a critical 1.65 to a robust 3.8. The elasto-plastic analysis demonstrated that yielding initiates at approximately 152 MPa, and exceeding this limit results in permanent plastic deformation and residual strain, proving that rigorous adherence to high safety factors and operating strictly within the linear elastic region are mandatory prerequisites for preventing catastrophic failure in manned deep-sea missions.

8. References

1. Young, W.C., "Roark's Formulas for Stress and Strain", McGraw-Hill.
2. Koç University, "Project 2 - MECH 303 Machine Design Guidelines", Fall 2025
3. ANSYS Inc., "ANSYS Mechanical User's Guide, Release 2025 R2".